Protein World-Model Building and Breaking with DAG+MDL Discovery

Automatically generated report

May 1, 2026

Abstract

This report documents an automatically generated protein world-model discovery run. The system studies real PDB structures, computes Gaussian Network Model normal-mode features from C-alpha contact networks, and fits compact DAG equations to normalized experimental B-factors. The run demonstrates a staged build-break-rebuild loop: a simple elastic-network model is built on compact proteins, challenged by terminal and hinge/domain-motion regimes, and revised only when Minimum Description Length (MDL) supports the added structure.

Contents

1 Purpose and Thesis **2**

2 Dataset and Physics Pipeline **2**

2.1 Core Variables 2

3 World-Building and Breaking Loop **2**

4 Quantitative Trajectory **2**

5 World-Model Evolution Figures **2**

6 Iteration-by-Iteration Scientific Interpretation **5**

6.1 Iteration 0: initial compact-protein stage 5

6.2 Iteration 1: stage_1_terminal_flexibility 5

6.3 Iteration 2: stage_2_hinge_domain_motion 5

6.4 Iteration 3: stage_3_validation_mixed 5

7 Protein Case Studies **5**

8 Final Conclusions **9**

A Appendix: Proteins Studied **10**

B Appendix: Highest B-Factor Residues **10**

B.1 1UBQ chain A 10

B.2 1CRN chain A 10

B.3 1AAR chain A 10

B.4 2CI2 chain I 10
B.5 4AKE chain A 11
B.6 1AKE chain A 11
B.7 1P38 chain A 11
B.8 1HEL chain A 11

1 Purpose and Thesis

The goal is not to claim a new law of protein dynamics. The goal is to demonstrate the central thesis of world-model discovery on real scientific data: an AI system should build an explicit model, seek regimes where it fails, revise the ontology, and ground each revision against physical measurements. Here the physical substrate is a Gaussian Network Model (GNM), and the measurement target is crystallographic B-factor variation within each protein chain.

2 Dataset and Physics Pipeline

Protein flexibility discovery from real PDB structures. Each residue is represented as a node in a C-alpha contact network. A Gaussian Network Model (GNM) computes normal-mode fluctuation features from the structure. The target is the residue's experimental crystallographic B-factor normalized within each protein chain. The Builder attempts to discover a compact explanatory law relating graph-normal-mode features, boundary features, and residue context to experimental flexibility. For each protein chain, residues are represented by C-alpha nodes. Edges connect nodes within 10.0 Angstrom. The GNM Kirchhoff matrix is diagonalized, and low-frequency modes are used to compute mean-square fluctuation features. The target, `bfactor_z`, is the per-chain z-score of C-alpha B-factor.

2.1 Core Variables

- `gnm_fluct_z`: normalized mean-square fluctuation from the GNM pseudoinverse.
- `gnm_fluct_log_z`: log-transformed fluctuation, useful when raw fluctuations are heavy-tailed.
- `terminal_exposure`: boundary/terminus exposure, highest at chain ends.
- `mode1_abs_z`: magnitude of participation in the slowest nonzero GNM mode.
- `hinge_score_z`: local curvature of the slowest mode, used as a hinge-like proxy.

3 World-Building and Breaking Loop

At each iteration, the Builder fits a compact DAG equation to the revealed residues. The Breaker reveals the next staged protein regime. The previous DAG is refit on the expanded data, then the search proposes candidate DAG edits. A new DAG is accepted only if its total description length is shorter: $L_{\text{total}} = L_{\text{model}} + L_{\text{data}}$.

4 Quantitative Trajectory

Iter	Revealed slice	Features	RMSE	R^2	L_{total}	Gain bits
0	initial	1	0.7194	0.482	236.0	0.0
1	stage_1_terminal_flexibility	2	0.5615	0.685	402.7	9.0
2	stage_2_hinge_domain_motion	2	0.6770	0.542	1094.5	37.3
3	stage_3_validation_mixed	1	0.7710	0.406	2016.5	54.3

5 World-Model Evolution Figures

The following automatically generated frames show the evolving equation, DAG structure, MDL terms, and fit against revealed observations.

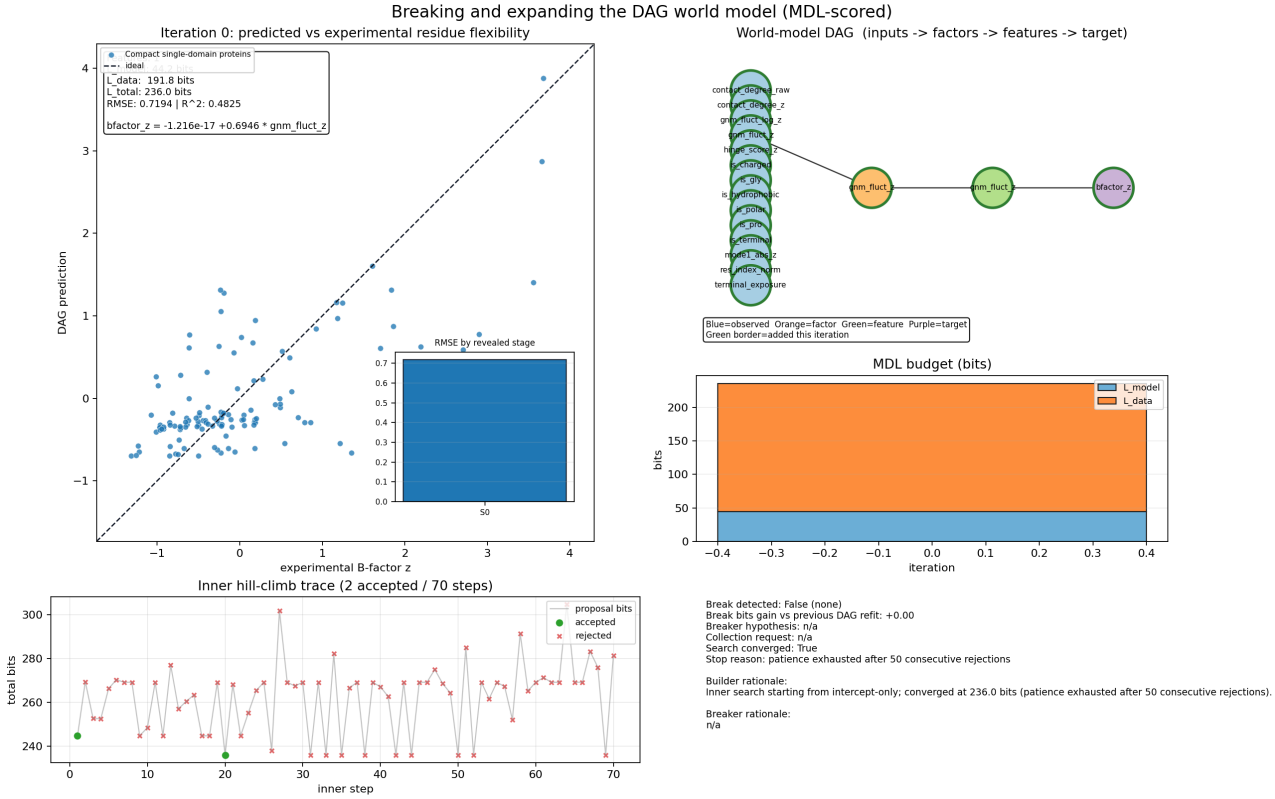


Figure 1: Frame Iter 00

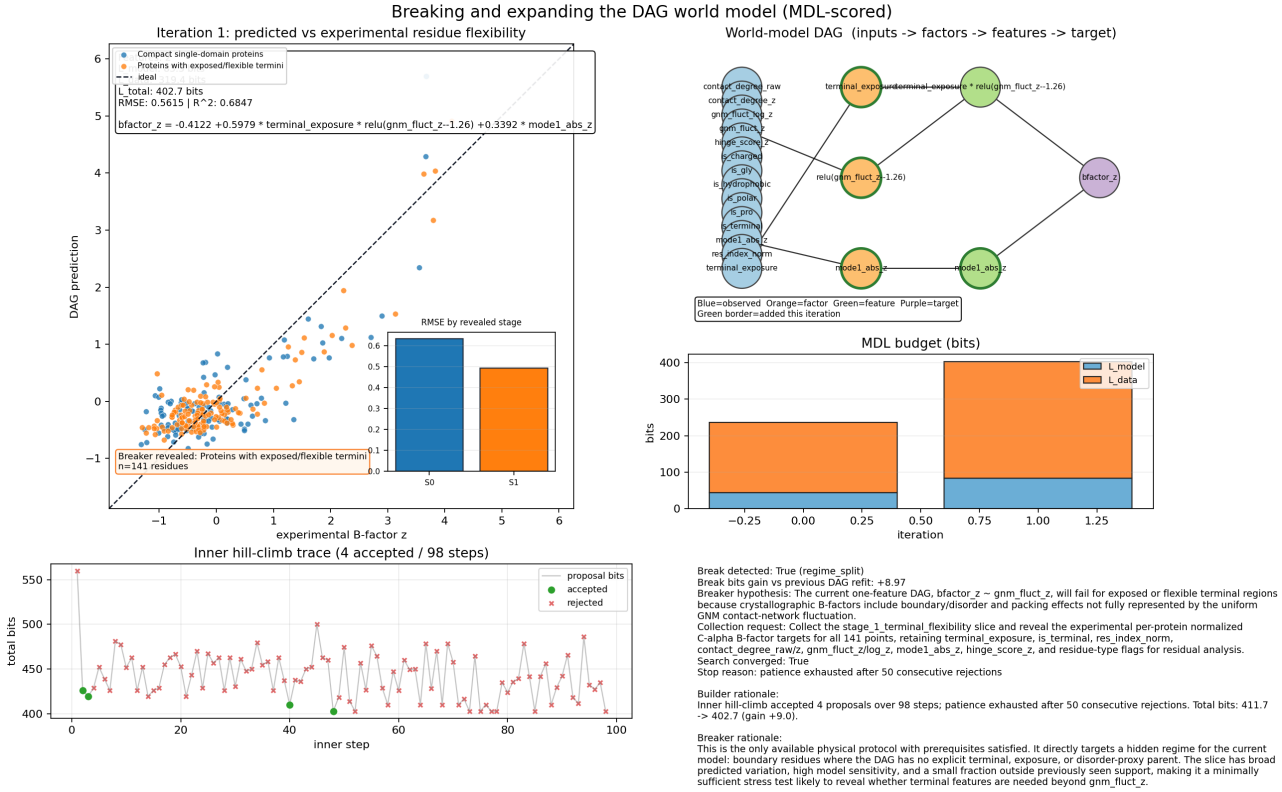
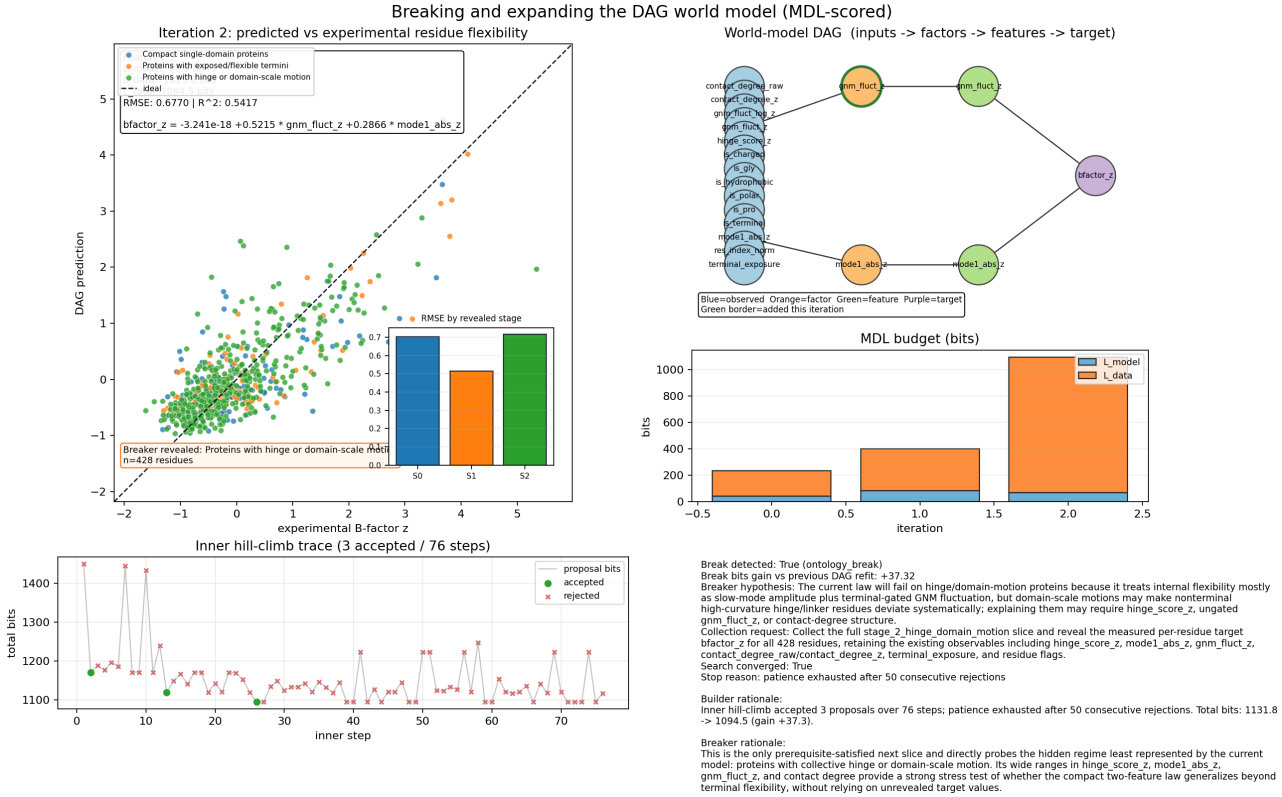


Figure 2: Frame Iter 01

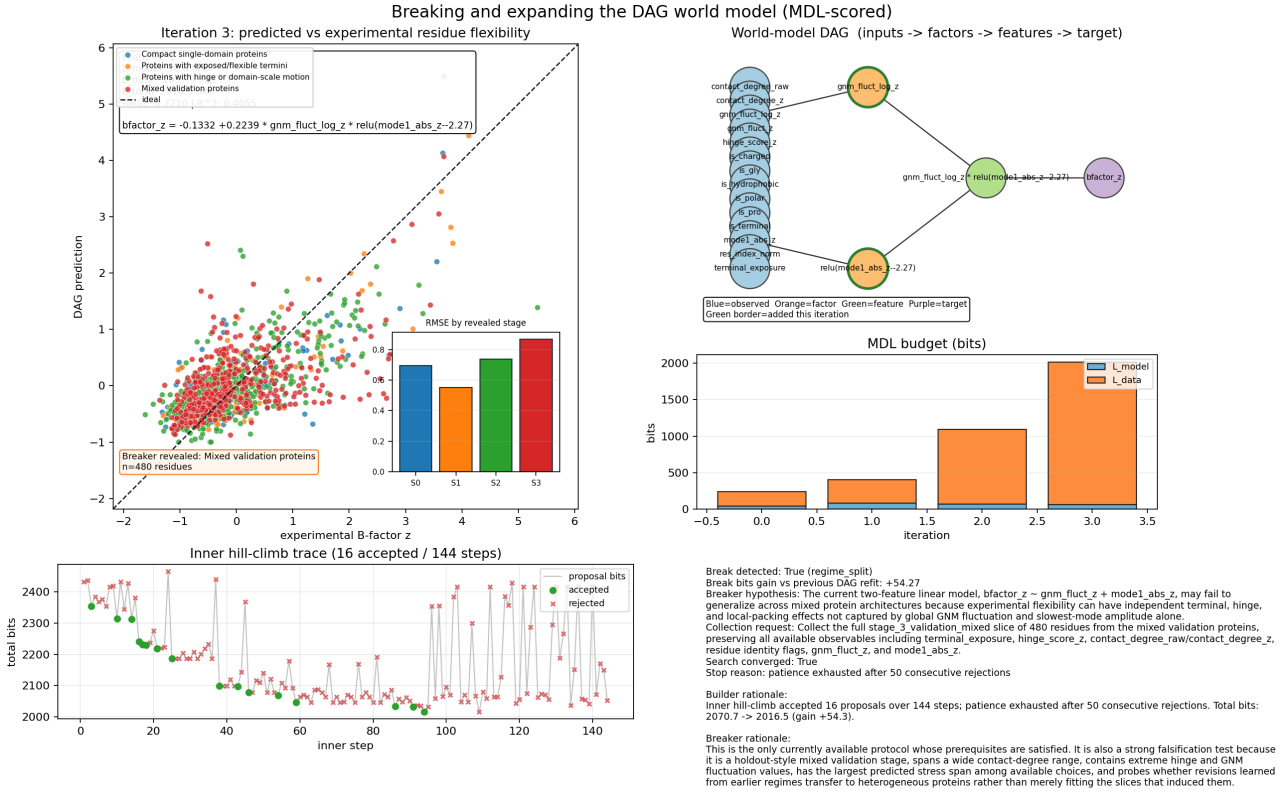


Break detected: True (ontology_break)
Break bits gain vs previous DAG refit: +37.32
Breaker hypothesis: The current law will fail on hinge/domain-motion proteins because it treats internal flexibility mostly as slow-mode amplitude plus terminal-gated GNM fluctuation, but domain-scale motions may make nonterminal high-curvature hinge/linker residues deviate systematically; explaining them may require hinge_score_z, ungated gnm_fluct_z, or contact-degree structure.
Collection request: Collect the full stage_2_hinge_domain_motion slice and reveal the measured per-residue target bfactor_z for all 428 residues, retaining the existing observables including hinge_score_z, mode1_abs_z, gnm_fluct_z, contact_degree_raw/contact_degree_z, terminal_exposure, and residue flags.
Search converged: True
Stop reason: patience exhausted after 50 consecutive rejections

Builder rationale:
Inner hill-climb accepted 3 proposals over 76 steps; patience exhausted after 50 consecutive rejections. Total bits: 1131.8 -> 1094.5 (gain +37.3).

Breaker rationale:
This is the only prerequisite-satisfied next slice and directly probes the hidden regime least represented by the current model: proteins with collective hinge or domain-scale motion. Its wide ranges in hinge_score_z, mode1_abs_z, gnm_fluct_z, and contact degree provide a strong stress test of whether the compact two-feature law generalizes beyond terminal flexibility, without relying on unrevealed target values.

Figure 3: Frame Iter 02



Break detected: True (regime_split)
Break bits gain vs previous DAG refit: +54.27
Breaker hypothesis: The current two-feature linear model, bfactor_z ~ gnm_fluct_z + mode1_abs_z, may fail to generalize across mixed protein architectures because experimental flexibility can have independent terminal, hinge, and local-packing effects not captured by global GNM fluctuation and slowest-mode amplitude alone.
Collection request: Collect the full stage_3_validation_mixed slice of 480 residues from the mixed validation proteins, preserving all available observables including terminal_exposure, hinge_score_z, contact_degree_raw/contact_degree_z, residue identity flags, gnm_fluct_z, and mode1_abs_z.
Search converged: True
Stop reason: patience exhausted after 50 consecutive rejections

Builder rationale:
Inner hill-climb accepted 16 proposals over 144 steps; patience exhausted after 50 consecutive rejections. Total bits: 2070.7 -> 2016.5 (gain +54.3).

Breaker rationale:
This is the only currently available protocol whose prerequisites are satisfied. It is also a strong falsification test because it is a holdout-style mixed validation stage, spans a wide contact-degree range, contains extreme hinge and GNM fluctuation values, has the largest predicted stress span among available choices, and probes whether revisions learned from earlier regimes transfer to heterogeneous proteins rather than merely fitting the slices that induced them.

Figure 4: Frame Iter 03

6 Iteration-by-Iteration Scientific Interpretation

6.1 Iteration 0: initial compact-protein stage

Accepted equation: `bfactor_z = -1.216e-17 +0.6946 * gnm_fluct_z` RMSE=0.7194, R^2=0.482, L_{total}=236.0 bits. Accepted DAG features: `gnm_fluct_z`.

- ‘gnm_fluct_z‘ is the direct GNM prediction: the residue is flexible because the C-alpha elastic network says it has a large mean-square fluctuation.

Scientific reading: the initial ontology is a classical elastic-network world. Residue flexibility is explained mainly by the contact graph's predicted GNM fluctuation.

6.2 Iteration 1: stage_1_terminal_flexibility

Accepted equation: `bfactor_z = -0.4122 +0.5979 * terminal_exposure * relu(gnm_fluct_z-1.26) +0.3392 * mode1_abs_z` RMSE=0.5615, R^2=0.685, L_{total}=402.7 bits. Accepted DAG features: `terminal_exposure * relu(gnm_fluct_z mode1_abs_z`. Break status: `regime_split`, detected=True, MDL gain=9.0 bits.

- ‘mode1_abs_z‘ adds global slow-mode participation. This points to collective domain-scale motion rather than only local packing.

Scientific reading: the model has shifted toward collective-motion ontology. Participation in the slowest global mode becomes part of the explanation.

6.3 Iteration 2: stage_2_hinge_domain_motion

Accepted equation: `bfactor_z = -3.241e-18 +0.5215 * gnm_fluct_z +0.2866 * mode1_abs_z` RMSE=0.6770, R^2=0.542, L_{total}=1094.5 bits. Accepted DAG features: `gnm_fluct_z`, `mode1_abs_z`. Break status: `ontology_break`, detected=True, MDL gain=37.3 bits.

- ‘gnm_fluct_z‘ is the direct GNM prediction: the residue is flexible because the C-alpha elastic network says it has a large mean-square fluctuation.
- ‘mode1_abs_z‘ adds global slow-mode participation. This points to collective domain-scale motion rather than only local packing.

Scientific reading: the model has shifted toward collective-motion ontology. Participation in the slowest global mode becomes part of the explanation.

6.4 Iteration 3: stage_3_validation_mixed

Accepted equation: `bfactor_z = -0.1332 +0.2239 * gnm_fluct_log_z * relu(mode1_abs_z-2.27)` RMSE=0.7710, R^2=0.405, L_{total}=2016.5 bits. Accepted DAG features: `gnm_fluct_log_z * relu(mode1_abs_z-2.27)`. Break status: `regime_split`, detected=True, MDL gain=54.3 bits.

- The retained DAG features should be read as the current compact hypothesis for normalized residue B-factor.

7 Protein Case Studies

Each figure shows the same C-alpha trace twice: left colored by normalized experimental B-factor, right colored by GNM-predicted fluctuation. Agreement indicates where the elastic-network world model is adequate; disagreement marks places where boundary, collective, crystallographic, or chemical context may be missing.

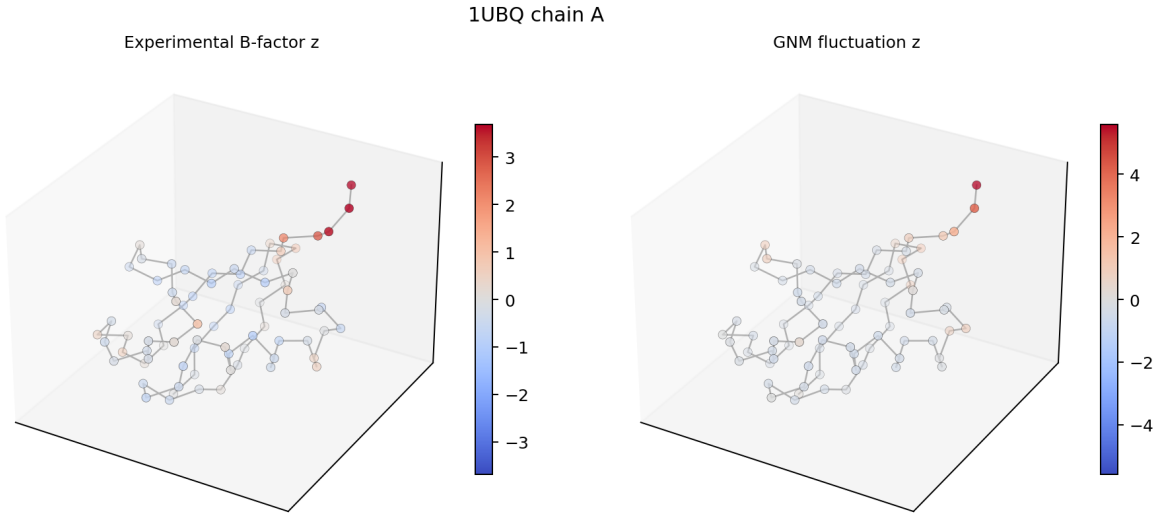


Figure 5: 1UBQ chain A in stage_0_compact_single_domain. n=76, GNM/B-factor correlation=0.807.

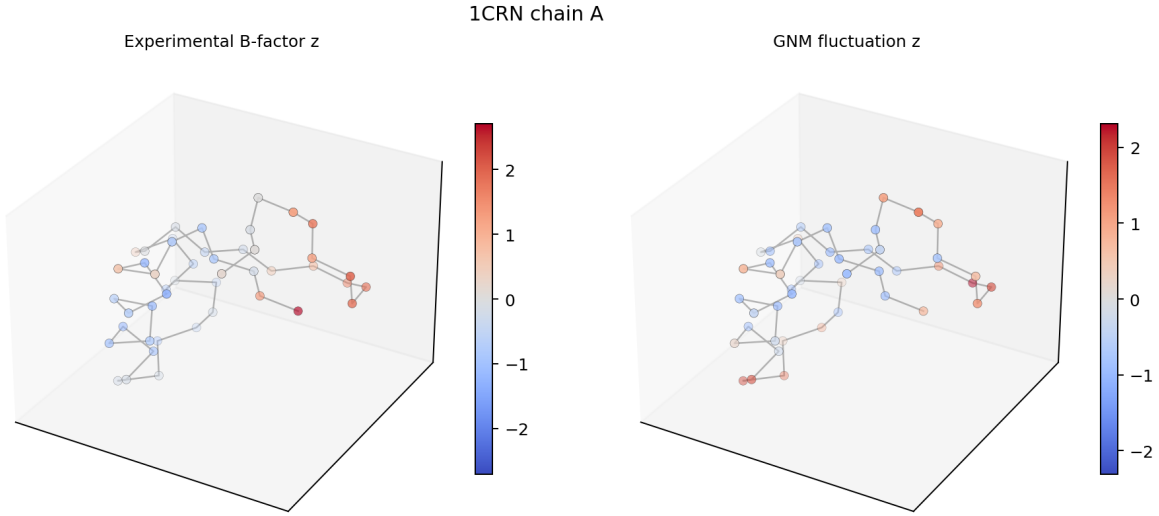


Figure 6: 1CRN chain A in stage_0_compact_single_domain. n=46, GNM/B-factor correlation=0.508.

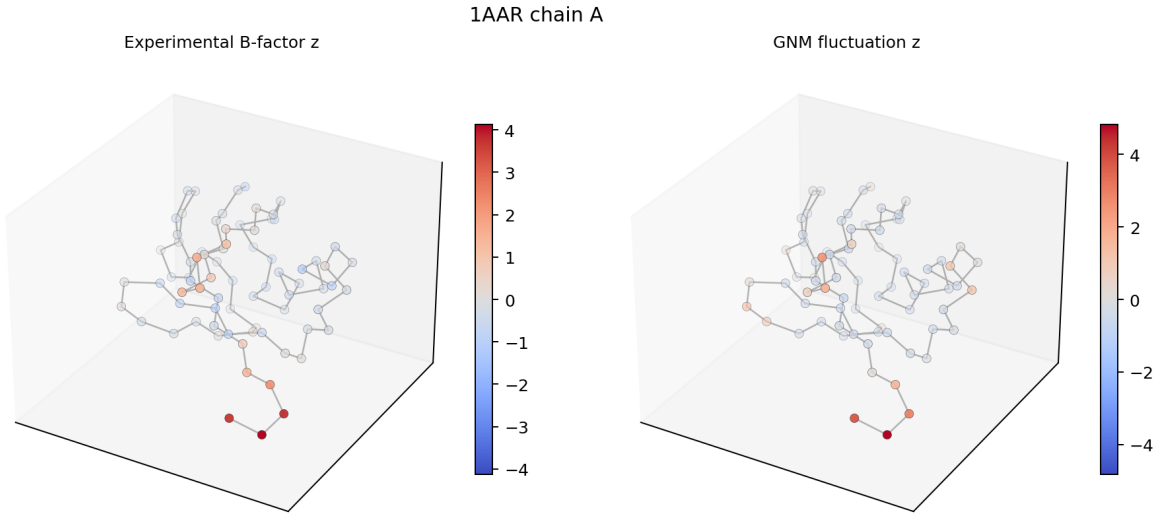


Figure 7: 1AAR chain A in stage_1_terminal_flexibility. n=76, GNM/B-factor correlation=0.837.

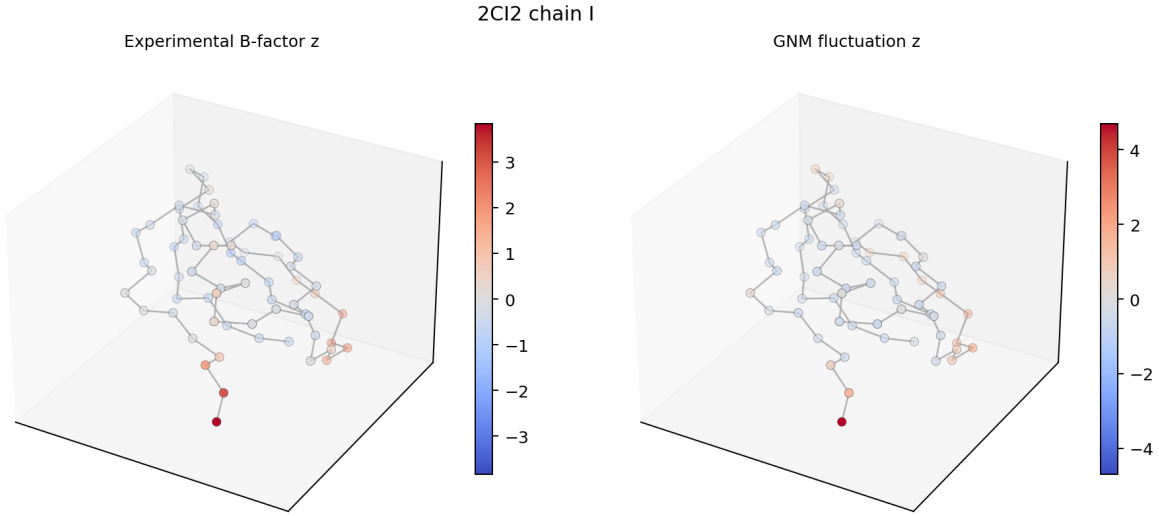


Figure 8: 2CI2 chain I in stage_1_terminal_flexibility. n=65, GNM/B-factor correlation=0.835.

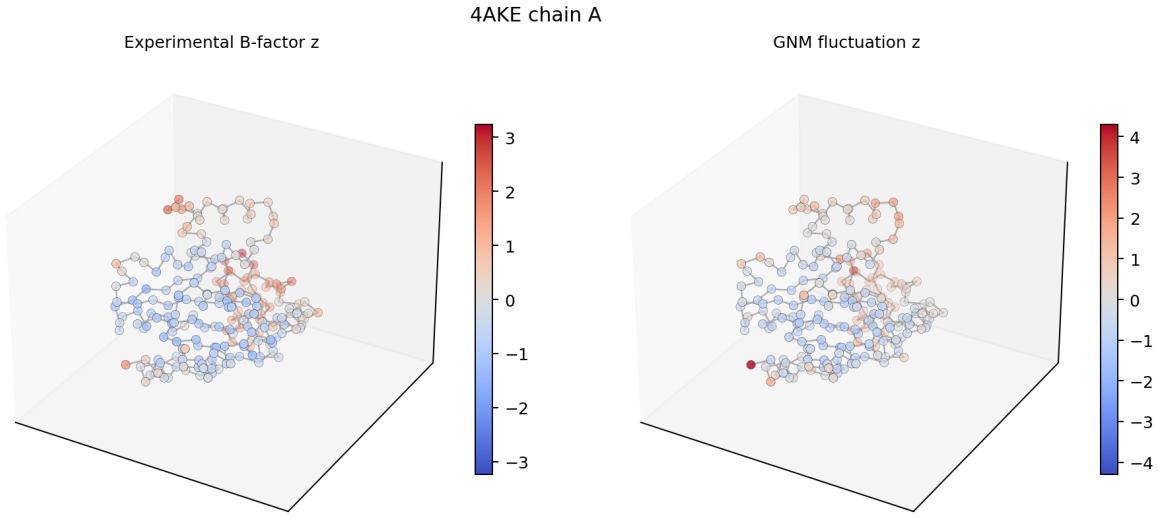


Figure 9: 4AKE chain A in stage_2_hinge_domain_motion. n=214, GNM/B-factor correlation=0.803.

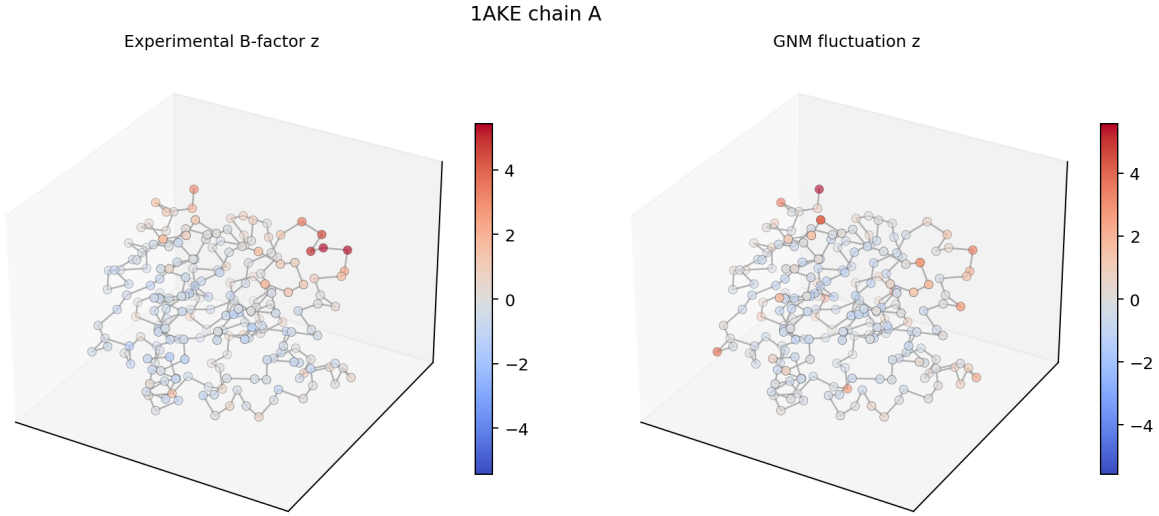


Figure 10: 1AKE chain A in stage_2_hinge_domain_motion. n=214, GNM/B-factor correlation=0.514.

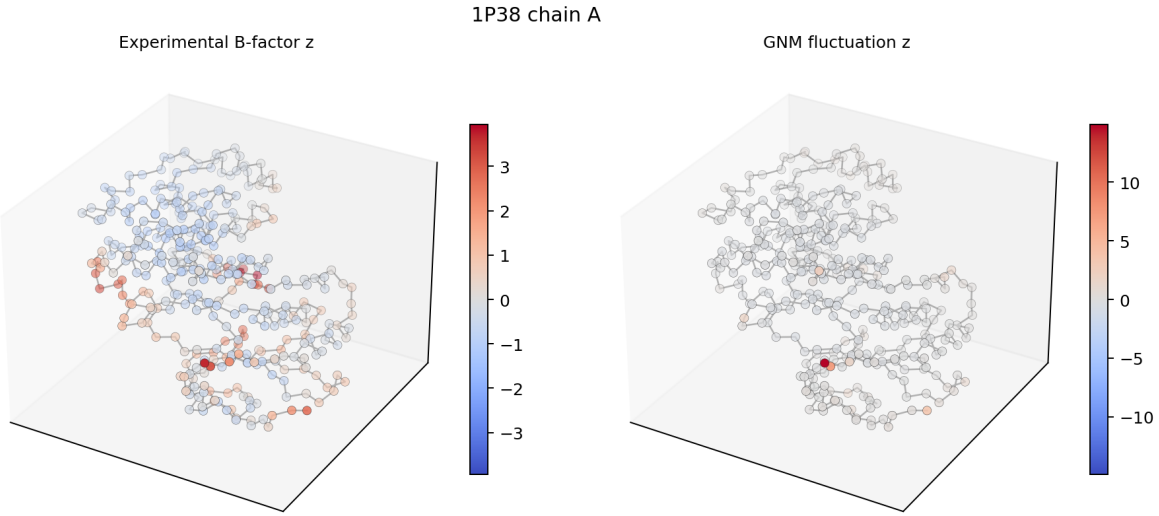


Figure 11: 1P38 chain A in stage_3_validation_mixed. n=351, GNM/B-factor correlation=0.427.

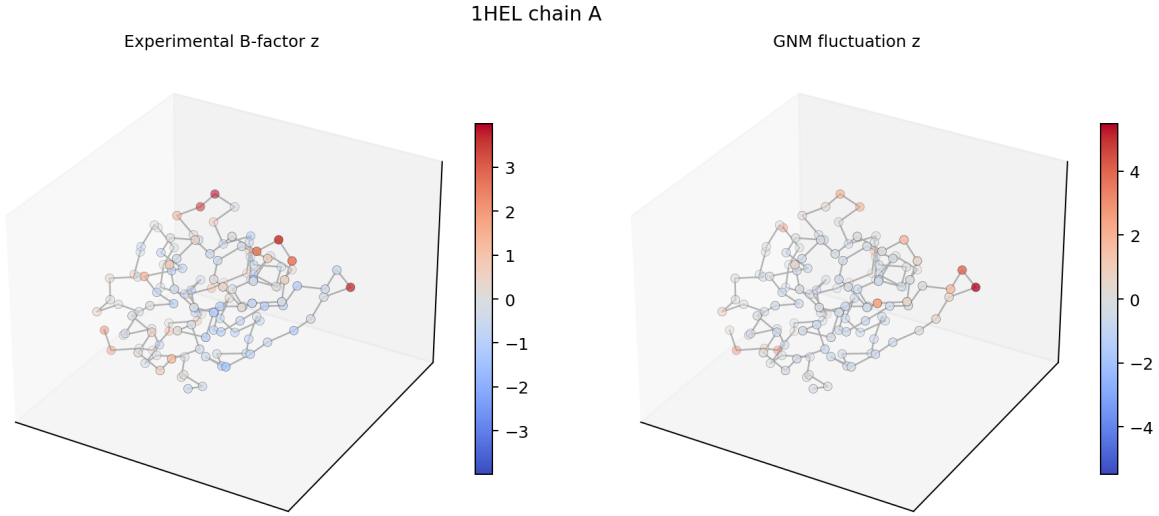


Figure 12: 1HEL chain A in stage_3_validation_mixed. n=129, GNM/B-factor correlation=0.481.

8 Final Conclusions

1. The first world model is physically meaningful: C-alpha contact-network normal modes explain a substantial fraction of residue-level B-factor variation in compact proteins.
2. The model breaks when new protein regimes expose systematic residual structure. In this run, accepted revisions involved boundary exposure, slow-mode participation, and log-scaled GNM fluctuation.
3. The accepted equations are not black boxes. Each feature is a mechanistic statement about contact topology, boundary position, or collective motion.
4. The result is a methodological discovery demonstration rather than a new biological law. A stronger scientific claim would require a larger benchmark, resolution controls, external validation, and manual structural interpretation of outliers.

A Appendix: Proteins Studied

PDB chain	Stage	Residues	Resolution	Corr(GNM,B)	Terminal frac
1UBQ:A	stage_0_compact_single_domain	76	2.00	0.807	0.132
1CRN:A	stage_0_compact_single_domain	46	2.00	0.508	0.217
1AAR:A	stage_1_terminal_flexibility	76	2.00	0.837	0.132
2CI2:I	stage_1_terminal_flexibility	65	2.00	0.835	0.154
4AKE:A	stage_2_hinge_domain_motion	214	2.00	0.803	0.047
1AKE:A	stage_2_hinge_domain_motion	214	2.00	0.514	0.047
1P38:A	stage_3_validation_mixed	351	2.00	0.427	0.028
1HEL:A	stage_3_validation_mixed	129	2.00	0.481	0.078

B Appendix: Highest B-Factor Residues

B.1 1UBQ chain A

Residue	B-factor z	GNM fluct z
A:76:G	3.68	5.58
A:75:G	3.66	4.13
A:74:R	3.56	2.03
A:73:L	2.90	1.12
A:72:R	2.19	0.90

B.2 1CRN chain A

Residue	B-factor z	GNM fluct z
A:1:T	2.70	0.85
A:39:T	1.98	0.65
A:38:A	1.86	1.26
A:37:G	1.84	1.89
A:41:P	1.71	0.88

B.3 1AAR chain A

Residue	B-factor z	GNM fluct z
A:75:G	4.12	4.81
A:74:R	3.80	2.96
A:76:G	3.63	3.71
A:73:L	2.23	1.62
A:10:G	1.54	2.46

B.4 2CI2 chain I

Residue	B-factor z	GNM fluct z
I:19:N	3.83	4.69
I:20:L	3.13	1.60
I:59:M	2.37	1.87
I:58:T	2.26	2.37
I:60:E	2.02	2.27

B.5 4AKE chain A

Residue	B-factor z	GNM fluct z
A:128:P	3.23	2.74
A:127:A	2.63	1.34
A:129:S	2.53	2.35
A:130:G	2.49	3.77
A:41:S	2.24	1.05

B.6 1AKE chain A

Residue	B-factor z	GNM fluct z
A:76:D	5.42	0.53
A:75:E	5.34	3.40
A:77:C	4.98	-0.15
A:78:R	4.43	1.18
A:79:N	3.46	0.70

B.7 1P38 chain A

Residue	B-factor z	GNM fluct z
A:174:H	3.94	0.79
A:175:T	3.87	0.22
A:173:R	3.71	-0.10
A:176:D	3.68	1.38
A:354:D	3.67	14.87

B.8 1HEL chain A

Residue	B-factor z	GNM fluct z
A:102:G	3.98	2.34
A:47:T	3.58	5.48
A:71:G	3.44	1.47
A:101:D	3.41	0.54
A:129:L	2.69	0.37